Counter-strike 2 game data collection with cheat labelling

Mille Mei Zhen Loo & Gert Luzkov

December 4, 2024

Abstract

In 2023 Valve launched Counter-Strike 2(CS2). Ever since the release of CS2 the game had problems with users deploying cheats to gain unfair advantages. It is known that machine learning can be used to classify cheaters, however this is a very data intensive task, and no dataset with cheater annotations exists for CS2. This paper aims to build a dataset with cheater labelling for the video game Counter-Strike 2. This was achieved by web scraping the CS2 match sharing site csstats.gg. This yielded XXXXX demofiles with the distribution of XXXX containing a VAC-banned user(cheater), and XXXX where no banned users were present. The quality of the data labels was determined by manual review of 100 demos where a user who was later banned was present, and 100 where no banned users were present.

Contents

1 Glossary

This paper does not include much hard to understand theory, however we do realise that not everyone

CS Counter-Strike. 4

CS2 Counter-Strike 2. 1, 3–5, 7, 9

CS:GO Counter-Strike: Global Offensive. 3, 5

CT counter-terrorists. 4

fps first-person shooter. 4

match sharing service A service provided by a third party, where users can upload their matches. The services often provide the means of analysing play styles with the intention of improving performance for the users.. 7

mod modification. 4

Overwatch In the context of Counter-Strike, overwatch refers to a match reviewing system, available in Counter-Strike: Global Offensive (CS:GO). This feature allowed trusted players to review demos with suspected cheaters. The investigators would then come to a verdict, and . Although there was no official route as to verify whether a This feature was removed 22/3/2023 as the Counter-Strike 2 (CS2) limited tests began. Version 1.38.5.5 was the last version of CS:GO, where this feature was available, and has as of this day not been reintroduced.[11][4][5][6]. 5, 7

T terrorists. 4

VAC Valve anti-cheat. 1

2 Introduction

2.1 First person shooters and cheating in video games

A first-person shooter (fps) is a video game genre centered around weapon based combat experienced from a first person perspective. These games can be played both offline and online, with the latter option being particularly popular due to it's ability to facilitate competitive and cooperative multiplayer experiences.

Due to the competitive nature of fps-games cheating is not an uncommon occurrence. However, the concept of cheating is often subjective, as players may hold varying perspectives on what constitutes cheating. While some players might view specific actions as unfair or illegitimate, others may consider them acceptable gameplay strategies. Despite the subjective aspects of cheating the paper *Cheating and anti-cheat system action impacts on user experience* by Bryan van de Ven identifies the three most common definitions of cheating in video games as the following:

"

1. *Using additional code or software to modify the game*
2. *Abusing faulty code in the game without using additional code or software.*
3. *Performing certain actions without using or altering code at all, such as looking at your friend's screen.*

"[10, ch. 2, p. 5]

For the purposes of this paper, cheating is defined as "*the use of additional code or software to modify the game*", as this is the most prevalent type of cheating occurring within the Counter-Strike community.

2.2 Counter-Strike

Counter-Strike (CS) is an online fps video game franchise, that started as a modification (mod) of the video game Half-Life[9][7]. In CS players join one of two teams: the terrorists (T) and the counter-terrorists (CT). The terrorists' objective is to plant a bomb at one of two designated bomb sites, labelled "A" and "B," while the counter-terrorists are tasked with defending these sites and defusing the bomb if planted. Victory conditions vary by team: the terrorists win if the bomb detonates or if all counter-terrorists players are eliminated, while the counter-terrorists succeed by defusing the bomb or eliminating all terrorists players. Throughout the game, performance in terms of kills, round-wins, and -losses yields in-game currency. This currency can be used to purchase armour, weapons, and utility at the start of each round. Hence, the game requires a precise aim, quick reflexes, economic strategy, and knowledge of the maps. This culminates in a very high skill ceiling.
In September 2023 a new version of the game Counter-Strike was published, namely CS2. Ever since the release of CS2, the game has had a problem with it's users deploying cheats to gain unfair advantages.
something the developers at Valve are aware of since the report button features several reasons for banning a player:

1.

CS2
Machine learning could potentially be used to detect such cheating behaviour. However, this requires a large amount of data, and no dataset with cheater annotations exists for CS2.

Creating a dataset for cheaters in CS2 is a challenging task, due to the complex nature of the data and lack of publicly available datasets on confirmed cheaters. There are multiple different types of cheating in an online first person shooter game such as CS2. Additionally, distinguishing between professional level players and experienced cheaters can be very difficult.

The goal of this paper is to create a labelled dataset on CS2 cheaters, as currently there is no publicly available labelled dataset. This data is essential for training machine learning models to detect cheating behaviours in real time. Additionally, the dynamic evolution of cheats makes it difficult to have a static, non-evolving dataset on cheaters, as cheats evolve over time as well, which in turn affects player behaviour.

A cheat detection machine learning model was created for the previous version of Counter-Strike, namely CS:GO. Unfortunately, the dataset used for training the model is not publicly available, and the methods of obtaining labelled data (Overwatch) are no longer available with the release of CS2[11][8][2]. Pro-player data is available through the site HLTV[3]. However, this dataset only contains the data of official professional level matches, hence no cheaters are present in this data.

In this project we aim to construct a labelled dataset of CS2 game data and dataset documentation.

3 Related work

this dataset was made with the intention of using it in a thesis
Without explicit approval from participants, we are unable to publish this dataset, without anonymising the dataset.

4 Methodology

4.1 Gathering data

In order to gather match data, one needs access to the data itself. This is challenging as matches that are played on the Valve server aren't shared by any official sources. Other works utilise access to match data via the Overwatch service provided by steam[2]. However, as of the launch of CS2 this feature is no longer accessible[8][11].

The problem was mitigated by web scraping a match sharing service. The platform csstats.gg has an all matches page which showcases the latest 50 matches that have been uploaded to the site.[1]

- csv
- pandas
- collections
- os
- cloudscraper
- re
- BeautifulSoup
- datetime
- pygame
- math

The gathering of data was done via the use of steam share codes.
other papers used overwatch + pro matches. Overwatch no longer exists. therefore other source is needed. enter csstats
what is csstats? maybe not this section
went to all matches. match by match in all matches went to a player, and took all codes within the last 30 days as this is the threshold, when valve deletes demos stored on their server

4.2 Manual review and data quality

In order to determine the quality of the data labels, a manual data review was conducted on a subsample of the dataset. This subsaple
how did we classify that a person was cheating
there are different ways of cheating

5 Data format and structure

5.1 Quality of data labels

Data is gathered in labelled .dem files, which are binary files for the CS2 game.

6 Preprocessing and cleaning

7 Rights statement

8 Related work

this dataset was made with the intention of using it in a thesis
Without explicit approval from participants, we are unable to publish this dataset, without anonymising the dataset.

References

[1] CSStats. *CS2 Stats Automatic stat and match tracking*. 2020. URL: https://csstats.gg/.

[2] Harry Dunham. *Cheat Detection using Machine Learning within Counter-Strike: Global OffensiveGlobal Offensive*. 2020. URL: https://openworks.wooster.edu/cgi/viewcontent.cgi?article=11803&context=independentstudy.

[3] HLTV. *Counter-Strike News & Coverage*. URL: https://www.hltv.org/. (accessed: 2.dec 2024).

[4] SteamDB. *Counter-Strike 2 (App 730) Patches and Updates*. URL: https://steamdb.info/app/730/patchnotes/. (accessed: 3.dec 2024).

[5] SteamDB. *Counter-Strike 2 update for 16 February 2023*. URL: https://steamdb.info/patchnotes/10562092/. (accessed: 3.dec 2024).

[6] SteamDB. *Release Notes for 3/22/2023*. URL: https://steamdb.info/patchnotes/10832117/. (accessed: 3.dec 2024).

[7] Valve. *Counter-Strike 2*. URL: https://www.counter-strike.net/. (accessed: 4.dec 2024).

[8] Valve. *Counter-Strike: Global Offensive Overwatch FAQ*. URL: https://blog.counter-strike.net/index.php/overwatch/. (accessed: 3.dec 2024).

[9] Valve. *Half-Life*. URL: https://www.half-life.com/en/halflife. (accessed: 4.dec 2024).

[10] Bryan van de Ven. *Cheating and anti-cheat system action impacts on user experience*. 2023. URL: https://www.cs.ru.nl/bachelors-theses/2023/Bryan_van_de_Ven___1024205___Cheating_and_anti-cheat_system_action_impacts_on_user_experience.pdf.

[11] Counter-Strike Wiki. *Overwatch*. URL: https://counterstrike.fandom.com/wiki/Overwatch. (accessed: 2.dec 2024).

A CS2 report pop-up

Below is a pop-up that appears when pressing report on a users profile within a CS2 game.

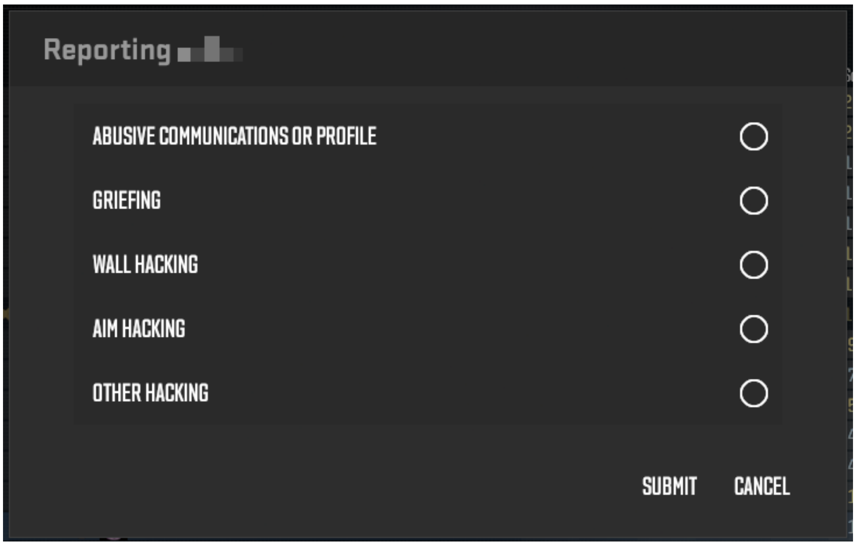

Figure 1: The popup shown when pressing report on a users account within Counter-Strike 2